CELL BIOLOGY / THE CELL CYCLE

How cells reproduce (mitosis)

One cell copies its DNA, then splits into two daughter cells that carry the same chromosomes.

GROW + COPY DNA

MITOSIS / THE NUCLEUS DIVIDES

THE CELL SPLITS

01



Interphase

The cell grows and copies its DNA: each chromosome becomes two sister chromatids.

02



Prophase

Chromosomes coil up into visible X shapes and the spindle begins to form.

03



Metaphase

Spindle fibers line the chromosomes up across the middle of the cell.

04



Anaphase

The sister chromatids are pulled apart toward opposite ends of the cell.

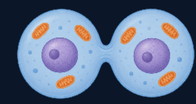
05



Telophase

A new nuclear envelope forms around each of the two sets of chromosomes.

06



Cytokinesis

The cytoplasm pinches in two, leaving two identical daughter cells.

each daughter cell can begin the cycle again at interphase

WHY IT MATTERS

Mitosis is how a body grows, replaces worn-out cells and heals wounds. When its checks fail, cells can divide out of control, as in cancer.

WHAT STAYS THE SAME

Each daughter cell gets a full copy of the DNA: the same chromosomes, in the same number, as the parent cell.

GROW / COPY / CONDENSE / ALIGN / SEPARATE / SPLIT

XYZ LAYOUT ENGINE

Simplified diagram: human body cells carry 46 chromosomes, not the few drawn here, and the phases blend into one another. Icons are hand-drawn vector art; the interphase and cytokinesis cells are AI-generated (GPT Image 2.5 via Higgsfield).

Reference: OpenStax, Biology 2e (2018), section 10.2 "The Cell Cycle", https://openstax.org/books/biology-2e/pages/10-2-the-cell-cycle